

Magnetoelastic wave in Ferromagnetic Thin-film Mediated by Dipolar Interaction

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Abstract. Magnetoelastic coupling mediated by magnetic dipolar interactions is theoretically investigated in ferromagnetic thin films under an in-plane magnetic field. We construct a microscopic framework that incorporates dipolar fields derived from Maxwell's equations in the presence of elastic deformation. The resulting coupled equations of motion predict hybridization between magnetostatic and Lamb waves. Numerical calculations for a yttrium iron garnet (YIG) film reveal hybridization of modes and anti-crossings of 0.1 to several MHz in the dispersion relations.

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Magnetoelastic coupling has attracted considerable attention because of its potential for manipulating magnetic properties via mechanical degrees of freedom [1, 2]. At the microscopic level, magnons and phonons are coupled through spin-orbit interaction, which gives rise to magnon-phonon coupling [1, 3, 4]. In contrast, at longer wavelengths the coupling is mainly mediated by the magnetic dipolar interaction. In this regime, magnetostatic waves interact with elastic waves. Experimental studies of such dipolar-interaction-mediated magnetoelastic coupling have recently progressed owing to the development of interdigital transducers (IDTs) [5, 6], which enable precise measurements of acoustic excitations in magnetic thin films. However, a theoretical framework that consistently incorporates the non-local nature of the dipolar interaction in magnetoelastic dynamics has not yet been established.

In this Letter, we construct a general formalism that incorporates the dipolar interaction and describe the resulting coupling between magnetostatic and elastic waves in ferromagnetic thin films.

Our system is schematically shown in Fig. 1. We consider a ferromagnetic thin film that is infinitely extended in the xy plane and has a finite thickness L in the z direction, with the origin taken at the middle of the film. The film is assumed to be isotropic both magnetically and elastically. An in-plane external magnetic field $\mathbf{H}_0$ is applied, resulting in an equilibrium saturation magnetization $\mathbf{M}_0$ parallel to $\mathbf{H}_0$.

We decompose the magnetic field $\mathbf{H}$ and the magnetization $\mathbf{M}$ as $\mathbf{H} = \mathbf{H}_0 + \mathbf{h}(\mathbf{r}, t)$ and $\mathbf{M} = \mathbf{M}_0 + \mathbf{m}(\mathbf{r}, t)$, respectively and study propagating magnetostatic waves of small deviations from the equilibrium magnetization $\mathbf{m}(\mathbf{r}, t)$. The system also supports elastic waves described by the displacement $\mathbf{u}(\mathbf{r}, t)$. The deformation modulates the distance between magnetic dipoles, so that the dipole field $\mathbf{h}(\mathbf{r}, t)$ generated by $\mathbf{m}$ is influenced by $\mathbf{u}$, leading to a magnetoelastic wave. We take the x axis parallel to the propagation direction, and assume that waves are plane waves and the system is uniform in the y direction. Then, $\mathbf{m}$ and $\mathbf{u}$ can be written as

$$\mathbf{m}(\mathbf{r}, t) = \mathbf{m}(z)e^{i(kx - \omega t)}, \quad \mathbf{u}(\mathbf{r}, t) = \mathbf{u}(z)e^{i(kx - \omega t)}. \quad (1)$$

We study the coupled dynamics of these waves up to first order in these quantities. Here, we focus exclusively on the case of an in-plane magnetic field.

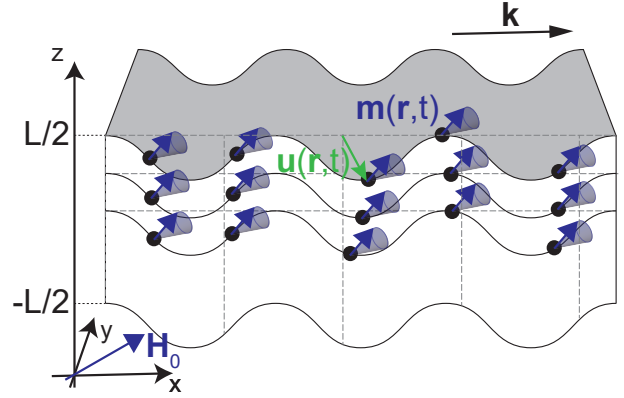


Figure 1. Schematic illustration of the system under consideration. A ferromagnetic thin film with thickness L in the z direction and infinite extent in the x and y directions is considered. Magnetic moments precess around an axis parallel to the in-plane external magnetic field $\mathbf{H}_0$. The elastic wave changes the positions of the magnetic moments and their separations, resulting in a modification of the dipolar interaction among them.

The case of an arbitrary external magnetic field will be presented elsewhere.

We start from the basic equations describing the magnetic dipolar interaction, namely Maxwell's equations. When elastic deformation is present, the magnetic dipoles deviate from their equilibrium positions $\mathbf{r}$ to $\mathbf{R} := \mathbf{r} + \mathbf{u}(\mathbf{r}, t)$. The Maxwell's equations are then rewritten in terms of this new coordinate as

$$\nabla_{\mathbf{R}} \cdot (\mathbf{H} + \mathbf{M}) = 0, \quad \nabla_{\mathbf{R}} \times \mathbf{H} \approx 0, \quad (2)$$

where in the second equation the magnetostatic approximation is applied and the time derivative of the electric field is discarded. This approximation is justified when the length scale is sufficiently long so that the dipolar interaction is dominant. By the plane-wave assumption (1), the dipolar field can also be written as $\mathbf{h} = \mathbf{h}(z)e^{i(kx - \omega t)}$. We rewrite Eq. (2) in terms of these quantities in the original coordinate $\mathbf{r}$. We note that we need to pay attention to the conservation of the magnetization in a unit volume [7] yielding $\mathbf{M}(\mathbf{R}) = \mathbf{M}(\mathbf{r})/\det(J(\mathbf{r}))$, where $J(\mathbf{r})$ is a Jacobian matrix (see Supplementary data [8] for details). We get

$$\nabla_{\mathbf{r}} \cdot (\mathbf{h}(\mathbf{r}) + \mathbf{m}(\mathbf{r})) = \mathbf{M}_0 \cdot \nabla_{\mathbf{r}} (\nabla_{\mathbf{r}} \cdot \mathbf{u}(\mathbf{r})), \quad (3)$$

$$\nabla_{\mathbf{r}} \times \mathbf{h}(\mathbf{r}) = 0, \quad (4)$$

where $\nabla_{\mathbf{r}}$ stands for derivatives with respect to the $\mathbf{r}$ coordinate. Because the surface of the film is also distorted as in Fig. 1, the boundary conditions used to solve these equations are also modified by the presence of $\mathbf{u}$ as

$$h_x^{\text{in}} - h_x^{\text{out}} = M_{0,z} \frac{\partial u_z}{\partial x}, \quad (5)$$

$$h_z^{\text{in}} + m_z - h_z^{\text{out}} = M_{0,x} \frac{\partial u_z}{\partial x}, \quad (6)$$

at $z = \pm L/2$, where $h^{\text{in/out}}$ stands for the dipolar fields inside and outside the film, respectively. Hereafter, lower indices indicate component in the $\mathbf{r}$ coordinate. Following the Green tensor method as in Ref. [9], the dipolar field satisfying Eqs. (3) and (4) under the boundary conditions (5) and (6) is given as (see Supplementary data [8] for details)

$$\mathbf{h}(z) = \int_{-\frac{L}{2}}^{\frac{L}{2}} dz' (G^m(z, z') \mathbf{m}(z') + G^u(z, z') \mathbf{u}(z')), \quad (7)$$

where

$$G^m(z, z') = \begin{pmatrix} -G_P(z, z') & 0 & -iG_Q(z, z') \\ 0 & 0 & 0 \\ -iG_Q(z, z') & 0 & G_{P'}(z, z') \end{pmatrix}, \quad (8)$$

$$G^u(z, z') = -ikM_{0,x}G^m(z, z'),$$

$$G_P(z, z') = \frac{k}{2} e^{-k|z-z'|}, \quad (9)$$

$$G_Q(z, z') = \text{sgn}(z - z') G_P(z, z'), \quad (10)$$

$$G_{P'}(z, z') = G_P(z, z') - \delta(z - z'). \quad (11)$$

This dipolar field couples the magnetostatic wave and the elastic wave in the film. We note that in the present setup there is no coupling when the equilibrium magnetization is parallel to the y axis, i.e., when the external magnetic field is applied in that direction. Otherwise, the two waves are coupled via $\mathbf{h}$.

Next, we examine the effect of magnetoelastic coupling on the dynamics of the magnetostatic and elastic waves. In the absence of this coupling, the dynamics of the magnetization $\mathbf{m}$ up to first order is described by the linearized Landau–Lifshitz equation

$$\dot{\mathbf{m}} = -\gamma\mu_0 (\mathbf{m} \times \mathbf{H}_0 + \mathbf{M}_0 \times \mathbf{h}), \quad (12)$$

where γ is the gyromagnetic ratio and μ_0 is the vacuum magnetic permeability. This equation can be derived from a Lagrangian

$$L^m[\mathbf{m}, \dot{\mathbf{m}}, \mathbf{m}^*, \dot{\mathbf{m}}^*] = \int dz \Re \left[\frac{1}{4M_0^2\gamma} \mathbf{M}_0 \cdot (\mathbf{m} \times \dot{\mathbf{m}}^*) - \frac{\mu_0}{4} \left(\mathbf{h}^* \cdot \mathbf{h} + \frac{H_0}{M_0} \mathbf{m}^* \cdot \mathbf{m} \right) \right]. \quad (13)$$

We can verify this by constructing the Euler–Lagrange equation with respect to $\mathbf{m}^*$ from this Lagrangian. On

the other hand, the dynamics of the displacement field $\mathbf{u}$ is described by the Navier–Cauchy equation

$$\rho \ddot{\mathbf{u}} = \mu \nabla^2 \mathbf{u} + (\mu + \lambda) \nabla (\nabla \cdot \mathbf{u}), \quad (14)$$

where ρ is the mass density, μ and λ are Lamé constants. The Lagrangian for this equation is

$$L^u[\mathbf{u}, \dot{\mathbf{u}}, \mathbf{u}^*, \dot{\mathbf{u}}^*] = \int dz \left[\frac{\rho}{4} \dot{\mathbf{u}}^* \cdot \dot{\mathbf{u}} + \frac{1}{4} \partial_i \sigma_{ij} u_j^* \right], \quad (15)$$

where $\sigma_{ij} = \mu(\partial_i u_j + \partial_j u_i) + \lambda \delta_{ij} \nabla \cdot \mathbf{u}$ is the stress tensor. The total Lagrangian of the coupled system is given by the sum of the two Lagrangians, Eqs. (13) and (15). In this case, $\mathbf{h}$ is no longer only a functional of the magnetization, but also a functional of the displacement field $\mathbf{u}$. As a result, a new volume force term $-\mu_0 \mathbf{f}$ appears in the Navier–Cauchy equation (14), where the force $\mathbf{f}$ is given by (see Supplementary data [8] for details of the derivation)

$$f_i(z) := \frac{\delta h_j^*}{\delta u_i^*} h_j = -ikM_{0,x} h_j(z). \quad (16)$$

Physically, this force can be understood as a Maxwell stress acting on the magnetic dipoles, caused by the change in the dipolar field induced by the displacement field. The coupled dynamics of the magnetoelastic wave is then given by

$$\begin{cases} \dot{\mathbf{m}} = -\gamma\mu_0 (\mathbf{m} \times \mathbf{H}_0 + \mathbf{M}_0 \times \mathbf{h}) \\ \rho \ddot{\mathbf{u}} = \mu \nabla^2 \mathbf{u} + (\mu + \lambda) \nabla (\nabla \cdot \mathbf{u}) - \mu_0 \mathbf{f}. \end{cases} \quad (17)$$

In order to examine the dynamical properties of the coupled waves described by the above equations of motion, we numerically calculate the dispersion relation in the following. Under the assumption (1), the time derivative on the left-hand side of the equations becomes $-i\omega$. The second equation of Eqs. (17) contains spatial derivatives of $\mathbf{u}$ and requires boundary conditions to solve it. Since we consider a film in vacuum, we impose the tension-free condition at the surfaces $z = \pm L/2$ as

$$\sigma_{xz}|_{z=\pm \frac{L}{2}} = \sigma_{yz}|_{z=\pm \frac{L}{2}} = \sigma_{zz}|_{z=\pm \frac{L}{2}} = 0. \quad (18)$$

Without coupling, it is known that a type of elastic wave called a Lamb wave appears under these conditions. Since the system in equilibrium is symmetric with respect to the $z = 0$ plane, the Lamb wave modes are either symmetric or antisymmetric with respect to the sign change of the variable z . Numerically, we need to solve the equations under the boundary conditions (18). For example, the finite element method naturally incorporates these conditions.

On the other hand, the magnetostatic waves without coupling to elastic waves in this geometry were studied in detail by Damon and Eshbach [10]. They analytically showed that when the external magnetic field is parallel to the propagation direction of the wave ($\mathbf{H}_0 \parallel \mathbf{k}$), the system has infinitely many bulk

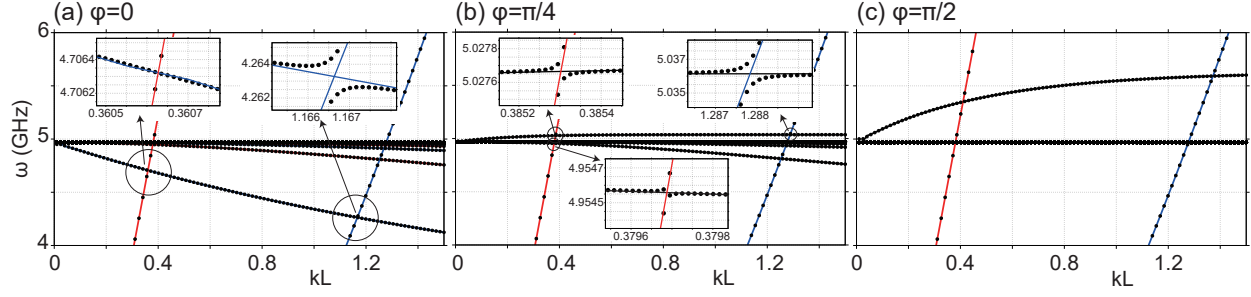


Figure 2. Dispersion curves of the coupled waves. Red and blue curves represent symmetric and antisymmetric modes of the uncoupled waves, including magnetostatic backward volume waves (BVWs) and Lamb waves, while black curves represent magnetostatic modes that are neither symmetric nor antisymmetric. Black dots show the dispersion of the coupled waves obtained from numerical calculations. The external magnetic field is applied (a) parallel to the propagation direction, $\mathbf{H}_0 \parallel \mathbf{k}$ ($\varphi = 0$), (b) at $\varphi = \pi/4$, and (c) perpendicular to the propagation direction, $\mathbf{H}_0 \perp \mathbf{k}$ ($\varphi = \pi/2$). Insets show magnified views of representative band crossings. In (a) $\varphi = 0$, anticrossings appear between BVW and Lamb modes only when they have the same C_{2x} eigenvalues (same color). When the C_{2x} symmetry of the system is broken (b), both the magnetostatic surface mode and the BVW couple to Lamb modes.

modes called backward volume waves (BVWs), which are either symmetric or antisymmetric. When the magnetic field is in-plane but perpendicular to the propagation direction ($\mathbf{H}_0 \perp \mathbf{k}$), a surface mode appears that is localized at one of the surfaces.

For the calculation of the dispersion, it is convenient to introduce another coordinate (r_1, r_2, r_3) , where the original coordinate (x, y, z) is rotated around the z axis so that the r_1 axis is parallel to the external magnetic field. Then, the r_1 component of $\mathbf{m}$ is zero since $\mathbf{M}_0 \cdot \dot{\mathbf{m}} = 0$ in the first equation of Eq. (17), and the equation is reduced to an equation for the two-component vector $(m_2, m_3)^T$.

Now we consider the coupled equations (17). Using the finite element method, we discretize the equations in real space while satisfying the boundary conditions, which leads to a generalized eigenvalue problem for the eigenvalue ω . By solving this problem, we obtain the dispersion curves shown in Fig. 2. The parameters are taken to be those of Yttrium-Iron-Garnet (YIG): $\rho = 5.17 \text{ g/cm}^3$, $\mu = 7.81 \times 10^{-8} \text{ kg} \cdot \text{GHz}^2/\text{m}$, $\lambda = 10.94 \times 10^{-8} \text{ kg} \cdot \text{GHz}^2/\text{m}$ [11], $\gamma = 29.8 \text{ GHz/T}$, and $\mu_0 M_0 = 0.178 \text{ T}$ [12]. The vacuum permeability is $\mu_0 = 4\pi \times 10^{-11} \text{ T}^2\text{m}/(\text{kg} \cdot \text{GHz}^2)$. We set the film thickness to $L = 500 \text{ } \mu\text{m}$ and the external magnetic field to $\mu_0 H_0 = 0.1 \text{ T}$.

We define the angle between the x axis and the external magnetic field as φ and write $\mathbf{H}_0 = H_0(\cos \varphi, \sin \varphi, 0)$. In Fig. 2(a), we plot the case $\varphi = 0$. In this case, magnetostatic BVWs appear, plotted as blue and red curves representing the antisymmetric and symmetric modes with respect to the twofold rotation about the x -axis (C_{2x}), with eigenvalues -1 and $+1$, respectively. Similarly, antisymmetric and symmetric Lamb waves with positive group velocities (positive slopes) are shown by blue and red curves,

respectively. The dots represent the dispersion of the coupled waves. The insets show magnified views of the band crossings between BVW and Lamb waves. We observe that the magnetostatic and elastic waves couple and produce anti-crossings of several MHz only when the two modes have the same C_{2x} eigenvalues. This is because when $\mathbf{H}_0 \parallel \mathbf{k}$, the system possesses the two-fold rotation symmetry with respect to the x axis C_{2x} and coupled modes needs to be either symmetric or anti-symmetric. Figure 2(b) shows the case $\varphi = \pi/4$. In this case, two types of magnetostatic modes coexist: BVWs and surface modes and the C_{2x} symmetry is broken. The insets show magnified views of the band crossings between the magnetostatic and Lamb modes. Even though the magnitude of the anti-crossings is small, couplings occur between all the modes as the two-fold rotation symmetry is broken in this case. Finally, in Fig. 2(c), we plot the case $\varphi = \pi/2$. In this case, a magnetostatic surface mode appears but does not couple to the Lamb modes. This is because $G^u = 0$, $\mathbf{f} = \mathbf{0}$ and Eq. (17) reduces to two independent equations. In other words, the magnetostatic modes are odd under the mirror reflection with respect to the xz plane, while the elastic modes are even, and therefore they do not couple.

In conclusion, we have constructed a theoretical framework to describe magnetoelastic coupling mediated by magnetic dipolar interactions in ferromagnetic thin films. The dipolar field was derived from Maxwell's equations in the presence of elastic deformations, and the coupled equations of motion were obtained within the Lagrangian formalism. Numerical calculations were performed for a YIG thin film to obtain the dispersion relations of the coupled modes. We found mode hybridizations and resulting anti-crossings of several MHz in general. When the external in-

plane magnetic field is parallel to the propagation of the waves, only magnetostatic and elastic waves with the same parity can hybridize. When they are perpendicular, couplings does not occur. The formulation developed here provides a starting point for theoretical studies of dipolar-interaction-mediated magnetoelastic coupling in ferromagnetic thin films.

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